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Composite Link Functions in Generalized Linear Models

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SUMMARY

In generalized linear models each observation is linked with a predicted value based on a linear function of some systematic effects. We sometimes require to link each observation with a linear function of more than one predicted value. We embed such models into the generalized linear model framework using composite link functions. The computer program GLIM-3 can be used to fit these models. Illustrative examples are given including a mixed-up contingency table and grouped normal data.

Keywords: GLIM; MIXED-UP TABLES; EM ALGORITHMS; GROUPED NORMAL DATA; RELATIVE RISK

1. INTRODUCTION

GENERALIZED linear models (Nelder and Wedderburn, 1972) have proved a useful unifying concept for identifying data structure and have allowed a flexible approach to model fitting. The generalized linear model is specified by

- (i) Independent observations $y_1, \dots, y_n$ distributed according to an exponential family distribution.
- (ii) A set of explanatory variables $x_{i1}, \dots, x_{ip}$, available on each observation ($i = 1, \dots, n$), describing the systematic linear component through $\eta_i = \sum x_{ij} \beta_j$, summing over j from 1 to p .
- (iii) The link function $g(\mu_i) = \eta_i$ relating the mean of an observation to the linear predictor η_i .

For example, consider a two-way contingency table with independence. The data, in the form of counts, can be thought of as arising from a Poisson distribution (for example, Birch, 1963). With independence, the mean of each observation (μ_i) can be thought of as a row effect times a column effect. This means that a logarithmic link function $\log(\mu_i) = \eta_i$ allows a linear structure to be defined linking the mean of an observation to explanatory variables representing rows and columns of the table.

Although generalized linear models are very flexible, some types of data do not fit directly into this framework. As an example consider the O, A, B system for human blood groups (Kempthorne, 1969, p. 172). Suppose that the respective gene frequencies are r , p and q and that there is random mating with respect to this locus. Then the frequencies in the four observed groups O, A, B and AB are r^2 , $p^2 + 2pr$, $q^2 + 2qr$ and $2pq$ respectively. One can think of the frequencies as arising from a "mixed-up" contingency table in which the AA, AO and OA cells are indistinguishable and similarly the BB, BO and OB cells, and the AB and BA cells are mixed-up.

If instead of linking each observation with one linear predictor we link a composite function of more than one linear predictor with each observation we can embed mixed-up contingency tables into the generalized linear model framework. In Section 2 we develop the necessary

Frequency table for O, A, B system

		A	O	B
		p	r	q
A	p	p^2	pr	pq
O	r	rp	r^2	rq
B	q	pq	qr	q^2

theory for such composite link functions. In Section 3 we show, using numerical examples, that this extension is useful not only for mixed-up contingency tables but in other contexts including grouped normal data and polychotomous responses. A further extension to the composite link function when the data can be usefully represented by the combination of two linear models is discussed briefly in Section 4. More general forms of composite link functions are considered in Section 5.

In Section 6 we discuss the efficiency of the procedure. In Section 7 some comments about computation are made including a GLIM-3 program for the contingency table example.

2. THEORY

The relationship between the mean of an observation and its linear predictor in generalized linear models can be written as $g(\mu_i) = \eta_i$ or alternatively $\mu_i = h(\eta_i)$. Sometimes we wish to associate more than one η_k (or correspondingly more than one set of systematic effects x_{kj} ($j = 1, \dots, p$)) with each observation y_i . We suppose that $\gamma_k = h(\eta_k)$ ($k = 1, \dots, m$), where m is the number of sets of systematic effects, and that $\mu_i = c_i(\gamma)$ where the c_i are known functions and γ is a $p \times 1$ vector with elements γ_k . Since each c_i can link μ_i , an expected value, with several η_k , functions of several linear predictors, we call c_i a *composite link function*. For ease of presentation we will now consider the special case when μ_i is a known linear function of γ and relegate the more general case to Section 5. Then μ_i can be written as $\mu_i = \sum c_{ik} \gamma_k$, summing over k from 1 to m . The model can be written as $E(\mathbf{y}) = \boldsymbol{\mu} = C\boldsymbol{\gamma}$, $\boldsymbol{\gamma} = h(\boldsymbol{\eta})$, $\boldsymbol{\eta} = X\boldsymbol{\beta}$, where $\boldsymbol{\mu}$, $\mathbf{y}$ are $n \times 1$ vectors, $\boldsymbol{\beta}$ is a $p \times 1$ vector and X is a $m \times p$ matrix with respective elements μ_i , y_i , β_i and x_{ij} . In this, C is a $n \times m$ matrix, the matrix analogue of the composite link functions.

Nelder and Wedderburn (1972) show that for their generalized linear models the solution of the likelihood equations for β_j ($j = 1, \dots, p$) is equivalent to an iterative weighted least-squares procedure. In each iteration a modified dependent variable given by $z_i = \eta_i + (y_i - \mu_i) \cdot (d\eta_i/d\mu_i)$ is used and the weight given to z_i is $w_i = (d\mu_i/d\eta_i)^2/\tau_i^2$ where $\tau^2 = d\mu/d\theta$ and θ is a parameter of the density function of y . For instance $\tau^2 = 1$ and μ for the Normal and Poisson distributions.

For our model with a composite matrix, C , a simple modification of Nelder and Wedderburn's iterative procedure gives maximum likelihood estimates of $\boldsymbol{\beta}$. In each iteration working dependent and independent variables $\mathbf{z} = CH\boldsymbol{\eta} + (\mathbf{y} - \boldsymbol{\mu})$ and $X^* = CHX$ are used in least-squares equations with weight function $1/\tau_i^2$ for z_i . Here, H is a $m \times m$ diagonal matrix with diagonal elements equal to $d\gamma_k/d\eta_k$. Obviously when $C = I$ and $\mu_k = \gamma_k$ this procedure is equivalent to that of Nelder and Wedderburn.

Since the inverse of a composite link function does not always exist we cannot, as Nelder and Wedderburn suggest for simple link functions, take the observations as initial estimates of the expected value (μ) and find estimates in the linear predictors ($\boldsymbol{\eta}$) through the inverse of the Ch . However, it is usually a simple matter to find satisfactory first estimates of $\boldsymbol{\eta}$ by using the values found from fitting a similar model with a simple link function.

3. EXAMPLES OF FITTING COMPOSITE LINK FUNCTION MODELS

We now illustrate the fitting of models incorporating composite link functions by considering three examples.

3.1. Mixed-up Contingency Tables

We take as an example the O, A, B system mentioned in the Introduction, with data given by Taylor and Prior (1938) and used by Kempthorne (1969). They observed 202, 179, 35 and 6 people in the 4 groups O, A, B and AB. The model fitted to the genotypes is additive on the logarithmic scale so that $h(\eta) = \exp(\eta)$. If we write $\mathbf{exp}(\boldsymbol{\eta})$ and $\mathbf{ln}(\boldsymbol{\eta})$ as the vectors with k th elements $\mathbf{exp}(\eta_k)$ and $\mathbf{ln}(\eta_k)$, the model fitted is of the form $\mu = \mathbf{Ch}(X\boldsymbol{\beta}) = \mathbf{Cexp}(X\boldsymbol{\beta})$ where X and $\boldsymbol{\beta}$ are respectively 9×4 and 4×1 matrices and C , the matrix analogue of the composite link function, is a 4×9 matrix. Write $\boldsymbol{\beta} = \mathbf{ln}(\boldsymbol{\alpha})$ where $\boldsymbol{\alpha}$ is a 4×1 vector given by $\boldsymbol{\alpha}^t = (N, p, r, q)^t$. Here, N is the total number of observations, and p, r, q are the frequencies of A, O and B.

The matrices X , C and y are given by

$$X^t = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 2 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

and $y^t = (179 \ 6 \ 202 \ 35)$.

In this example, following Birch (1963), we think of the observations as coming from independent Poisson distributions so that $\tau_i^2 = \mu_i$; also since $h = \exp$, $d\gamma_k/d\eta_k = \exp(\eta_k)$. There are dependencies in the columns of the design matrix X and just as in ordinary linear models we need to introduce constraints. We use the constraint $\beta_1 = 0$.

Using initial estimates of $\beta_j = \frac{1}{2} \ln(422/9)$ ($j = 2, 3, 4$) convergence was attained in 5 iterations. The fitted values for the O, A, B and AB were 205.9, 175.0, 30.6 and 10.6 respectively. Hence estimates of p, r and q are 0.252, 0.698 and 0.050. The deviance is 3.17 with one degree of freedom. The GLIM-3 coding for this example is given in Section 7.

3.2. Grouped Normal Data

The exact analysis of grouped normal data has been presented by (among others) Sampford (1952). We apply the method to data taken from Snedecor and Cochran (1967, p. 105) which refer to gains in weight of two samples of female rats under two diets. For the purposes of this example we round the data to the nearest 10 g so that they now appear as

	Diet 1							
Weight gain	0-85	-95	-105	-115	-125	-135	-145	-155
Frequency	1	0	2	2	3	2	0	1

	Diet 2						
Weight gain	0-75	-85	-95	-105	-115	-125	125-
Frequency	1	0	2	1	1	1	1

The original data are assumed to be normally distributed with means μ_1 and μ_2 for the two diet levels and a common variance σ^2 . Hence for each level the observed frequencies (y_i) are

model form $E(\mathbf{y}) = \boldsymbol{\mu} = Ch(\boldsymbol{\eta})$ with $\mathbf{y}^t = (43 \ 10 \ 24 \ \dots \ 19 \ 8 \ 11)^t$. We find

$$C = \begin{bmatrix} I_{10} & O_{10,10} & O_{10,10} \\ O_{10,10} & I_{10} & B_1 \end{bmatrix}, \quad X = \begin{bmatrix} I_{10} & O_{10,2} \\ I_{10} & B_2 \\ I_{10} & B_3 \end{bmatrix},$$

where I_t is an identity matrix of order t while B_1 is a diagonal matrix with i th element x_i ,

$$B_2^t = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^t$$

and

$$B_3^t = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}^t.$$

The function $h = \exp$ and $\boldsymbol{\beta}^t = (\ln p_1, \dots, \ln p_{10}, \ln \alpha_1, \ln \alpha_2)^t$. The observed number of controls may be used as initial estimates of p_i while a linear regression of the ratio cases/controls on x provides initial estimates of α_1 and α_2 . Assuming the 20 cell entries are Poisson variables then we obtain $\hat{p} = 0.00243$ with standard error 0.00073, the fitted values being given in Table 1.

TABLE 1

Dust exposure in lung cancer cases and controls, relative risks and fitted values from a linear relative risk model

Total dust exposure in units of [(million par- ticles per cubic foot) $\times$ (years)] up to 7 years before death of the case	Observed			Fitted		
	Cases	Controls	Relative risk	Cases	Controls	Relative risk
< 6,	43	285	1	43.5	284.5	1.00
≥ 6 , < 10	10	62	1.07	9.7	62.3	1.02
≥ 10 , < 30	24	166	0.96	25.7	167.3	1.02
≥ 30 , < 100	37	211	1.16	35.0	213.0	1.07
≥ 100 , < 300	31	168	1.22	31.8	167.2	1.24
≥ 300 , < 600	27	95	1.88	25.5	96.5	1.73
≥ 600 , < 1000	18	50	2.39	18.6	49.5	2.46
≥ 1000 , < 1500	10	19	3.49	10.0	19.0	3.43
≥ 1500 , < 2000	6	8	4.97	5.8	8.1	4.65
≥ 2000	9	11	5.42	9.5	10.5	5.87

4. BILINEAR MODELS

Consideration of the estimation of N_1 and N_2 in the grouped data example suggests an immediate generalization of the composite linear model. We suppose a model where

$$\eta_i = \sum_{j=1}^p \beta_j x_{ij}, \quad \gamma_k = h(\eta_k), \quad \eta_{1k} = \sum_{j=1}^{p_1} \beta_{1j} C_{kj} \gamma \quad \text{and} \quad \mu_k = h_1(\eta_{1k}).$$

This is an extension of the composite model in two ways. Firstly, the $n \times m$ matrix C has been replaced by a linear function of parameters β_{1j} i.e. $C = \sum \beta_{1j} C_j$, summing over j from 1 to p_1 where C_j are known matrices. Secondly, instead of letting $\mu = \eta_1$ we allow $\boldsymbol{\mu}$ to be some function, h_1 , of $\boldsymbol{\eta}_1$. We call this model a generalized bilinear model as two separate linear models are used to specify the model.

The grouped normal data example fits this bilinear framework with $\beta_{11} = N_1$, $\beta_{12} = N_2$,

$$C_1 = \begin{bmatrix} E_1 & O_{9,7} \\ O_{7,9} & O_{7,7} \end{bmatrix}, \quad C_2 = \begin{bmatrix} O_{9,9} & O_{9,7} \\ O_{7,9} & E_2 \end{bmatrix}$$

and h_1 equal to the identity function.

Again Nelder and Wedderburn's argument can be extended to show that the solution of the likelihood equations for β_1 and β is equivalent to an iterative weighted least squares scheme with dependent variate $\eta_1 + CH\eta + H_1(y - \mu)$, independent variates $C_j\gamma$ and CHX where H and H_1 are diagonal $m \times m$ and $n \times n$ matrices with diagonal elements equal to $d\gamma_k/d\eta_k$ and $d\eta_{1k}/d\mu_k$, and with weights $(d\mu_i/d\eta_{1i})^2/\tau_i^2$.

By applying the method of this section to the grouped normal example it can be found that the *ML* estimates of N_i are given by the sample sizes and that these estimates are uncorrelated with the estimates of the other parameters. In two other cases, truncated grouped normal data and Wadley's problem in bioassay (Finney, 1971, pp. 202–208) bilinear models are appropriate, but the sample sizes N_i are not known. The procedure in this section is then a convenient way of estimating all the parameters.

5. GENERAL COMPOSITE LINK FUNCTIONS

In Section 2 we discussed the case when μ is a linear function of γ . A similar procedure can be used if $\mu_i = c_i(\gamma)$. The working independent and dependent variables are now

$$x_{ij}^* = \sum_{k=1}^m (d\mu_i/d\eta_k) x_{kj} \quad (j = 1, \dots, p) \quad \text{and} \quad z_i = \sum_{k=1}^m (d\mu_i/d\eta_k) (y_i - \mu_i)$$

with weight function $1/\tau_i^2$. Also for the bilinear model in Section 4 when

$$\eta_{1k} = \sum_{j=1}^{p_1} \beta_{1j} c_{jk}(\gamma) \quad \text{and} \quad \mu_k = h_1(\eta_{1k}),$$

the working independent variables are

$$c_{ji}(\gamma) \quad (j = 1, \dots, p) \quad \text{and} \quad \sum_{k=1}^m (d\eta_{1i}/d\eta_k) x_{kj} \quad (j = 1, \dots, p),$$

and the working dependent variate is

$$z_i = \eta_{1i} + \sum_{k=1}^m (d\eta_{1i}/d\eta_k) \eta_k + (y_i - \mu_i) (d\eta_{1i}/d\mu_i)$$

with weight $(d\mu_i/d\eta_{1i})^2/\tau_i^2$.

6. EFFICIENCY

The composite link functions have been presented because they allow more models to be embedded simply into the generalized linear model framework. Of course in specific cases other procedures might be more efficient computationally. The *EM* algorithm (Dempster *et al.*, 1977) is a natural competitor. This method, for our models, uses the predicted value $\mu = C\gamma$ but uses the first differentials of the likelihood to give new estimates. This, we expect, can lead to less computation in each iteration but needs more iterations for convergence. Recently Wolynetz (1979) found this to be true for censored and grouped normal data.

7. COMPUTATION

The calculations involved can be easily programmed. It has been found quite simple to perform the calculations using GENSTAT (Nelder *et al.*, 1975) making use of the facilities for matrix operations and fitting linear models. Another possibility is to use GLIM-3 (Baker and Nelder, 1978) which was especially written for fitting generalized linear models.

Using GLIM, the general form of the analysis for any data will be: (1) Input of data, (2) Calculation of values needed in the analysis, (3) Declaration of macros, (4) Declaration of model, (5) Fitting of model, (6) Output of results.

These steps will be described in detail for the example in Section 3.1 (see Baker and Nelder, 1978 for details of the language, etc.):

```

$UNITS 4 $DATA Y $READ 179 6 202 35
$DATA 9 XP $READ 2 1 1 1 0 0 1 0 0
$DATA 9 XQ $READ 0 1 0 1 2 1 0 1 0
$DATA 9 XR $READ 0 0 1 0 0 1 1 1 2
$VARIATE 9 ETA GAM
$VARIATE 36 C $DATA 9 T
$READ 1 2 4 12 16 23 33 35 36
$CALCULATE C = 0 : C(T) = 1
$CALCULATE %L = %CU(Y) : %L = %LOG(%L)
$MACRO M1 $CAL %A = %NE(%PL,0)
$SWI% A MEXT
$CAL ETA = %PE(1)*XP+%PE(2)*XR
          +%PE(3)*XQ
$CAL GAM = %EXP(ETA) $CAL %N = 4
$WHI %N MAT $ENDMAC
$MACRO M2 $CAL %DR = 1 $ENDMAC
$MACRO M3 $CAL %VA = %FV $ENDMAC
$MACRO M4
$CAL %DI = 2*(%YV-%LOG(%YV/%FV)
          +%FV-%YV)
$ENDMAC
$MACRO MEXT $EXTRACT %PE $ENDMAC
$MAC MAT $CAL %N = %N-1 : %I = 4-%N
$CAL T = C(%GL(9,1)+9*(%I-1))*GAM
$CAL %FV(%I) = %CU(T)
$CAL %LP(%I) = %CU(T*ETA)
$CAL P(%I) = %CU(T*XP)
$CAL Q(%I) = %CU(T*XQ)
$CAL R(%I) = %CU(T*XR) $ENDMAC
$YVA Y $OWN M1 M2 M3 M4 $SCA 1
$VAR 3 %PE $CAL %PE = 1/2*%L-%LOG(3)
$CAL %LP = P = Q = R = 0
$FIT P+R+Q-%GM
$DISPLAY E V
$C STEP 1 READ IN
$C OBSERVATIONS
$C AND COLUMNS
$C OF X
$C STEP 2 DECLARE
$C TEMPORARY VECTORS
$C CONSTRUCT C (AS A
$C VARIATE OF LENGTH 36) AND
$C STORE LOG OF OBSERVATIONS
$C STEP 3 USING OWN FACILITY
$C TO GIVE NON-STANDARD
$C GLM,
$C FOUR MACROS
$C M1,M2,M3,M4
$C ARE NEEDED
$C MACRO M1 USES
$C 2 MACROS
$C MEXT AND MAT
$C STEP 4 DECLARES MODEL
$C AND INITIAL VALUES
$C STEP 5 FITS MODEL
$C 6 OUTPUT OF RESULTS

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